

C Programming

Data structure

Stack and Queues

Exercise01

How to use malloc

- Please create a program for the malloc usage example shown in the preparation video and check the execution result.
 - <https://wsdmoodle.waseda.jp/mod/quiz/view.php?id=5822909>

Example of using the malloc() function

```
#include<stdio.h>
#include<stdlib.h>

int main(void){
    int N; → int N, i;
    int *x;

    printf("Input N:");
    scanf("%d",&N);

    x = (int *) malloc(sizeof(int)*N);
    if(x==NULL){
        printf("Can't allocate memory.\n");
        exit(1);
    }

    for(i=0;i<N;i++) x[i]=i; // The same as the array of N double variables.

    for(i=0;i<N;i++) printf("x[%d]=%d\n",i,x[i]);

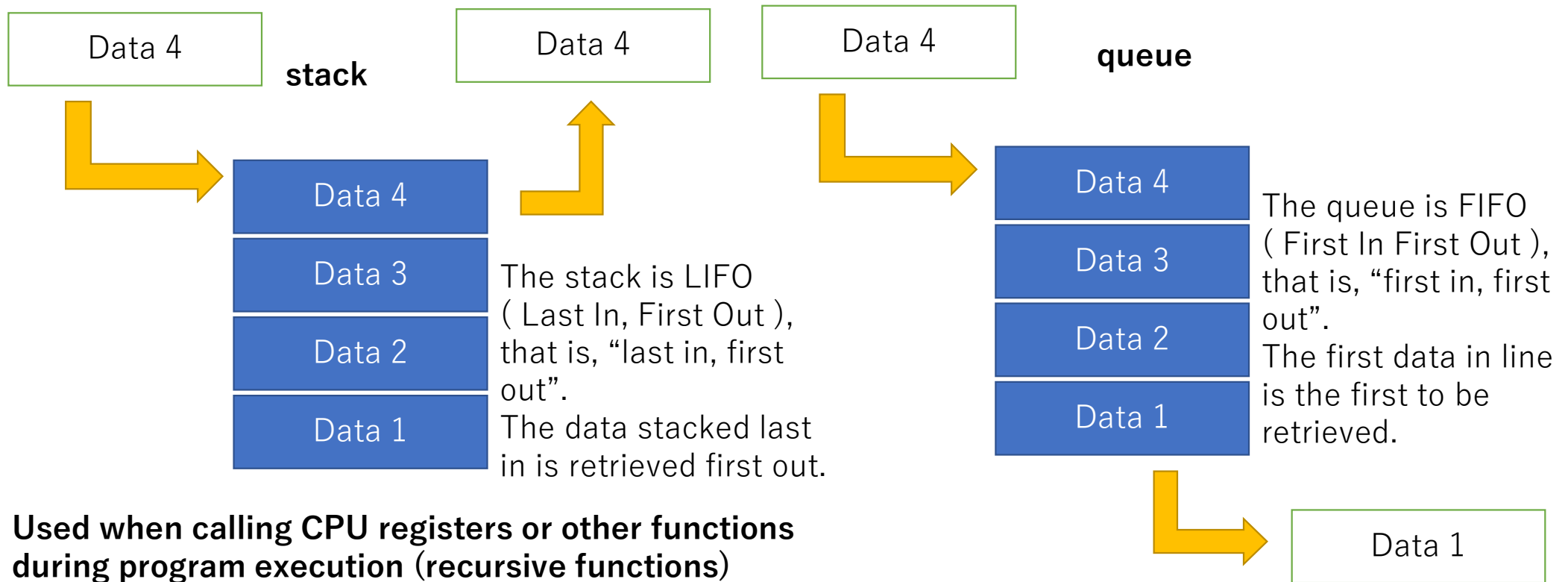
    free(x);
    return 0;
}
```

malloc and calloc

- malloc
 - Allocates a single block of requested memory
 - Clear and do not initialize allocated memory
 - `void *malloc(size_t size);`
- calloc
 - Allocate multiple blocks of requested memory
 - Initialize allocated memory to zero
 - If it is an integer, the value is guaranteed to be 0.
 - `void *calloc(size_t count, size_t size);`
- realloc
 - Functions for reallocating and allocating memory
 - `void *realloc(void *ptr, size_t size);`
 - `void *a = realloc(NULL, 1);`
 - `void *b = realloc(a, 2);`

Equivalent to <code>malloc(1)</code> ' (memory area allocated)
Expanding the memory space

Stack queue differences



<https://itmanabi.com/queue-stack/#toc6>
<https://suwaru.tokyo/%E3%80%90%E3%83%87%E3%83%BC%E3%82%BF%E6%A7%8B%E9%80%A0%E3%80%91%E3%82%AD%E3%83%A5%E3%83%BC%E3%81%A8%E3%82%B9%E3%82%BF%E3%83%83%E3%82%AF%E3%81%AE%E4%BD%BF%E3%81%84%E6%96%B9%E3%82%92%E8%A7%A3%E8%AA%AC/>

Can be used to implement "breadth-first search" and "cache"

Exercise02: Writing a Program for the Stack

Write a program to do the stack

1, Please define a structure

```
struct stack
{
    int head
    int arr[MAX_DATA];
};
```

Exercise02: Writing a Program for the Stack

2. Please push the following numbers in order.

[10, 15, 18, 12, 7, 9, 2, 14]

3. Please pop all the numbers.

4. Please implement POP processing and push processing in functions.

Exercise 03: Queued programs

1. Create a program to do the queuing.
2. Create a program to enqueue [10, 15, 18, 12, 7, 9, 2, 14] in order to an empty list.
3. Modify the program created in step 2 and add a process to dequeue until the list is empty.
4. * Please output the results of both enqueue and dequeue so that you can see the results on the way.

Please complete Exercises 1 through 3.

Exercise 04

Create a program that manages an integer array dynamically using malloc. The array should be able to expand and shrink based on user input using realloc.

- Implement the following operations:
 - add X – Add integer X to the array.
 - remove – Remove the last element from the array.
 - display – Show the current array contents.
 - exit – Terminate the program and free allocated memory.
- Input:
 - An initial size N (integer).
 - A series of commands (add X, remove, display, exit).
- Output:
 - Print the current state of the array after each operation.
- Constraints:
 - N is a positive integer.
 - Use realloc() to dynamically manage memory.
 - Ensure proper memory deallocation with free().

Exercise 05

- Problem:
 - Given a string S containing parentheses (), {}, [], determine if all parentheses are properly paired and correctly nested. Implement a program to verify this.
- Input:
 - A single string S (maximum length 100).
- Output:
 - If the parentheses are correctly placed, output:
 - Output 1: The parentheses are correctly placed.
 - Output 2: The parentheses placement is incorrect.
- Constraints:
 - S consists of alphanumeric characters and (), {}, [] parentheses only.
 - An empty string (S="") is considered valid.

Example Input 1:

```
{[()]}
```

Example Output 1:

```
The parentheses are correctly placed.
```

Example Input 2:

```
{( [ ] ) }
```

Example Output 2:

```
The parentheses placement is incorrect.
```

Hint:

A stack can be used to solve this problem by pushing opening brackets ((, {, [) onto the stack and popping when encountering closing brackets (), },]). If the stack is empty at the end, the parentheses are correctly placed.

Next week Online Classroom

- Next week's Topic: Report 2
- Online
 - Zoom
 - Please check Moodle for links, etc.